

INTERNAL IMPLEMENTATION PLAYBOOK

ChurchFlow Architecture & Delivery Guide

The Complete Technical, Pitching & Client Onboarding Cheatsheet

Core Architecture & Technology Stack Breakdown:

1. Meta WhatsApp Cloud API

• Enterprise WhatsApp Messaging

Handles Webhooks from incoming chats, interactive button replies, media delivery (flyers), and verified template broadcasts directly to phone numbers.

2. n8n Workflow Automation

• The Central Orchestration Engine

Parses webhooks, executes instant keyword routing (hymns, directions, bank info), feeds structured prompt context to OpenAI, and runs CRON broadcast triggers.

3. Supabase (PostgreSQL + Storage)

• Database & Real-Time Sync

Stores members, conversation threads, individual messages, and prayer requests. Hosts flyers in public buckets and pushes real-time WebSocket events to dashboard.

4. Next.js 16 Web Dashboard

• Staff Admin Command Center

Modern React dashboard with real-time WhatsApp-styled chat, Needs Pastor filtering, flyer uploaders, member directories, and role-based permissions.

5. OpenAI Intelligence (gpt-5.4-mini)

• Conversational Ministry AI

Provides context-aware spiritual responses, program queries, and answers general church questions while adhering to strict church doctrinal guidelines.

Message Routing Logic & Live Pitch Demo Script

1. How the Message Routing Engine Operates (Step-by-Step):

- Step A: Member sends message on WhatsApp "" Meta Cloud API fires a POST Webhook to n8n.
- Step B: n8n "Parse Incoming Message" extracts phone number, sender name, text, and button IDs.
- Step C: n8n logs the incoming message into Supabase "messages" table for record-keeping.
- Step D: Fast Direct Router checks for specific high-frequency queries (Bypassing AI):
- Hymns & Anthems: Exact verses & refrain from rccg-hymns.json database.
 - Directions / Address: Returns exact address + Google Maps navigation link.
 - Giving / Accounts: Returns Access Bank & UBA account numbers.
 - Service Times: Returns formatted weekly schedule + next upcoming service flyer.
- Step E: If not a direct match, "Build AI Context" packages upcoming events, knowledge base, and prompt "" OpenAI generates a spiritually grounded WhatsApp reply.
- Step F: Supabase Realtime alerts ChurchFlow Web Dashboard "" Populates live thread & triggers chime sound.

2. Live Demo Script (What to Show in 5 Minutes):

Follow this exact script when demoing to a Pastor or Committee to create immediate excitement:

1	Start with the Greeting Action: Send "Hi" from your phone to the bot. Outcome: Show them how the official interactive buttons instantly pop up on WhatsApp with Zero delay.
2	Ask for a Hymn Action: Type: "Please help me with blessed assurance hymn" Outcome: Show them how it outputs the complete verses, refrain, and scripture immediately.
3	Ask for Church Location Action: Type: "Where is the church located?" Outcome: Show the exact address and the clickable Google Maps navigation link.
4	Trigger Pastoral Escalation Action: Type: "I need urgent counseling from the pastor" Outcome: Switch screen to ChurchFlow Dashboard. Point out the glowing red Ø=P" Needs Pastor badge, sound chime, and live chat thread.
5	Reply from Dashboard Action: Type a message from the Web Dashboard Live Reply Box. Outcome: Show it arriving instantly on the phone with verified double checkmarks.

Objection Handling, Pricing & Client Onboarding

Handling Common Church Objections:

Q: "Will AI give incorrect spiritual answers or hallucinate?"

A: No. ChurchFlow is constrained with strict doctrinal prompts, verified knowledge base articles, and direct database lookups for hymns, accounts, and events.

Q: "Can our WhatsApp number get banned by Meta?"

A: No. ChurchFlow uses official Meta WhatsApp Cloud API credentials, adhering to official Meta Business terms, unlike unofficial bulk spam software.

Q: "What happens if a member needs human pastoral attention?"

A: The system instantly flags the conversation as "Needs Pastor" on the web dashboard with sound alerts and desktop notifications so pastors can take over.

Client Onboarding & Setup Checklist:

1. Get Phone Number: Purchase fresh SIM not tied to personal WhatsApp (WhatsApp Business Cloud requirement).
2. Meta Developer Setup: Register WhatsApp App on developers.facebook.com & generate System User Token.
3. Supabase Setup: Run database migrations (schema.sql), create storage buckets (Flyers), and set public read policies.
4. n8n Import: Import n8n-workflow-churchflow-fixed.json, configure environment credentials (Supabase & OpenAI).
5. Vercel Deployment: Connect GitHub repo to Vercel and input NEXT_PUBLIC_SUPABASE_URL and ANON_KEY.
6. Data Population: Ingest church schedule, bank accounts, flyers, and staff login accounts in Settings.

Recommended Pricing & Packaging Model:

- Setup & Implementation Fee: One-time setup for Meta Business verification, system integration, flyer uploads & staff training.
- Monthly Retainer / Maintenance: Covers server hosting, OpenAI token allowances, broadcast scheduling, flyer design updates, and technical support.
- High ROI Pitch: Replaces traditional bulk SMS costs while increasing first-timer retention and giving revenue.

